

Electronics 1: Homework #6

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Problem 4.15

For the NMOS common-source amplifier in Figure P4.15, the transistor parameters are: $V_{TN} = 0.8$ V, $K_n = 1$ mA/V², and $\lambda = 0$. The circuit parameters are $V_{DD} = 5$ V, $R_S = 1$ k Ω , $R_D = 4$ k Ω , $R_1 = 225$ k Ω , and $R_2 = 175$ k Ω . (a) Calculate the quiescent values I_{DQ} and V_{DSQ} . (b) Determine the small-signal voltage gain for $R_L = \infty$. (c) Determine the value of R_L that will reduce the small-signal voltage gain to 75 percent of the value found in part (b).

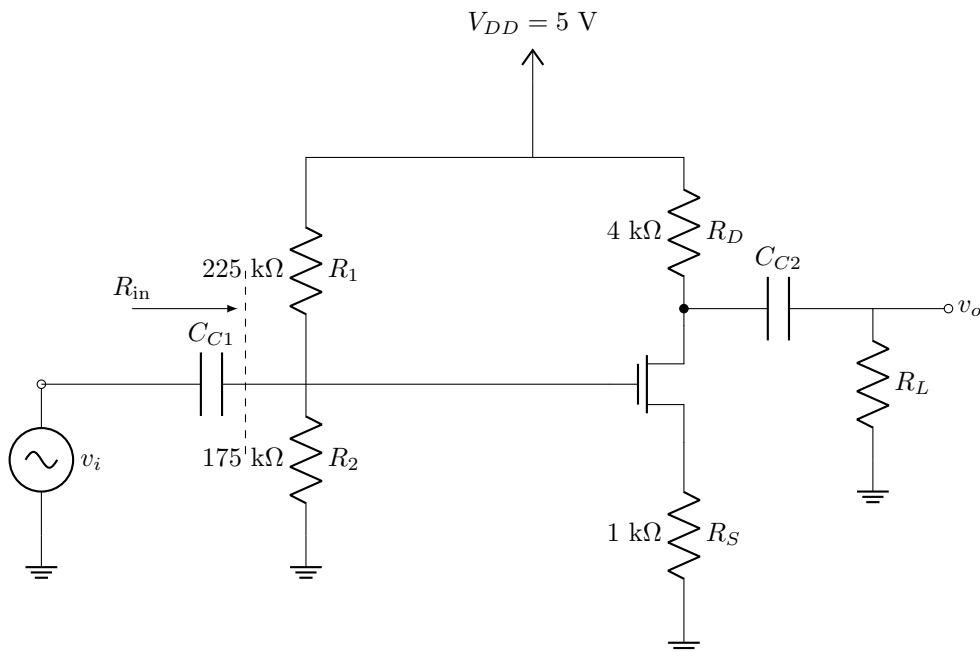


Figure 1: Figure P4.15

Solution

(a) Quiescent values I_{DQ} and V_{DSQ} :

$$V_G = \left(\frac{R_2}{R_1 + R_2} \right) V_{DD} = \left(\frac{175}{175 + 225} \right) (5) = 2.1875 \text{ V}$$

$$V_G = V_{GS} + I_{DQ}R_S = V_{GS} + K_n(V_{GS} - V_{TN})^2 R_S$$

$$2.1875 = V_{GS} + (1)(1)(V_{GS} - 0.8)^2 = V_{GS} + V_{GS}^2 - 1.6V_{GS} + 0.64$$

$$0 = V_{GS}^2 - 0.6V_{GS} - 1.5475 \Rightarrow V_{GS} = 1.58 \text{ V}$$

$$I_{DQ} = K_n(V_{GS} - V_{TN})^2 = (1)(1.58 - 0.8)^2 = 0.608 \text{ mA}$$

$$V_{DSQ} = V_{DD} - I_{DQ}(R_S + R_D) = 5 - (0.608)(1 + 4) = 1.96 \text{ V}$$

(b) Small-signal voltage gain for $R_L = \infty$:

$$g_m = 2\sqrt{K_n I_{DQ}} = 2\sqrt{(1)(0.608)} = 1.56 \text{ mA/V}$$

$$A_v = \frac{-g_m R_D}{1 + g_m R_S} = \frac{-(1.56)(4)}{1 + (1.56)(1)} = -2.44$$

(c) Value of R_L that reduces small-signal voltage gain to 75 percent:

$$\begin{aligned}A_v &= \frac{-g_m(R_D \parallel R_L)}{1 + g_m R_S} = \frac{-(1.56)(R_D \parallel R_L)}{1 + (1.56)(1)} = -0.6094(R_D \parallel R_L) \\-(0.75)(2.44) &= -0.6094(R_D \parallel R_L) \Rightarrow R_D \parallel R_L = 3.0 \text{ k}\Omega \\4 \parallel R_L &= 3 \Rightarrow \frac{1}{R_L} = \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \Rightarrow R_L = 12 \text{ k}\Omega\end{aligned}$$

Problem 4.18

The ac equivalent circuit of a common-source amplifier is shown in Figure P4.18. The small-signal parameters of the transistor are $g_m = 2 \text{ mA/V}$ and $r_o = \infty$. (a) The voltage gain is found to be $A_v = V_o/V_i = -15$ with $R_S = 0$. What is the value of R_D ? (b) A source resistor R_S is inserted. Assuming the transistor parameters do not change, what is the value of R_S if the voltage gain is reduced to $A_v = -5$.

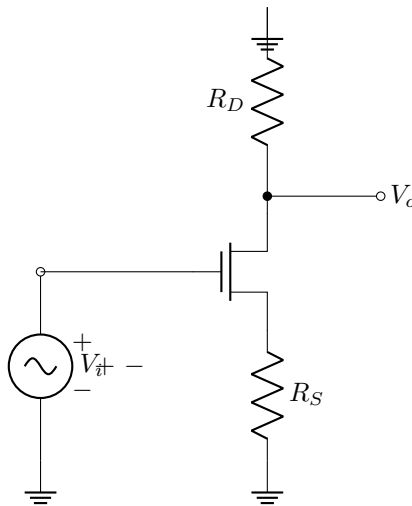


Figure 2: Figure P4.18

Solution

(a) Find the value of R_D for $A_v = -15$ and $R_S = 0$:

$$A_v = -g_m R_D$$

$$-15 = -(2)R_D \Rightarrow R_D = 7.5 \text{ k}\Omega$$

(b) Find the value of R_S if the voltage gain is reduced to $A_v = -5$:

$$A_v = \frac{-g_m R_D}{1 + g_m R_S}$$

$$-5 = \frac{-(2)(7.5)}{1 + (2)R_S}$$

$$1 + 2R_S = \frac{15}{5} = 3 \Rightarrow 2R_S = 2 \Rightarrow R_S = 1 \text{ k}\Omega$$

Problem 4.27

For the common-source amplifier shown in Figure P4.27, the transistor parameters are $V_{TP} = -1.2$ V, $K_p = 2$ mA/V², and $\lambda = 0.03$ V⁻¹. The drain resistor is $R_D = 4$ k Ω . (a) Determine I_Q such that $V_{SDQ} = 5$ V. (b) Find the small-signal voltage gain for $R_L = \infty$. (c) Repeat part (b) for $R_L = 8$ k Ω .

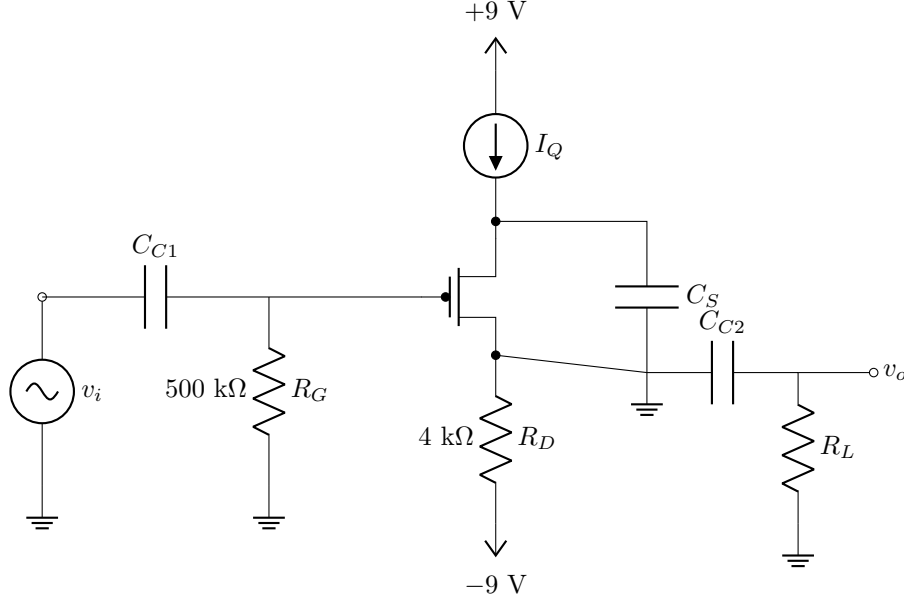


Figure 3: Figure P4.27

Solution

(a) Determine I_Q such that $V_{SDQ} = 5$ V:

$$\begin{aligned}
 V_S &= V_{SGQ} \\
 V_D &= I_{DQ}R_D - 9 \\
 V_{SDQ} &= V_S - V_D = V_{SGQ} - (I_{DQ}R_D - 9) \Rightarrow V_{SGQ} = V_{SDQ} + I_{DQ}R_D - 9 \\
 V_{SGQ} &= 5 + I_{DQ}(4) - 9 = 4I_{DQ} - 4 \\
 I_{DQ} &= K_p(V_{SGQ} + V_{TP})^2 = 2(V_{SGQ} - 1.2)^2 \\
 V_{SGQ} &= 4[2(V_{SGQ} - 1.2)^2] - 4 = 8(V_{SGQ}^2 - 2.4V_{SGQ} + 1.44) - 4 \\
 0 &= 8V_{SGQ}^2 - 20.2V_{SGQ} + 7.52 \Rightarrow V_{SGQ} = 2.071 \text{ V} \\
 I_{DQ} &= I_Q = K_p(V_{SGQ} + V_{TP})^2 = 2(2.071 - 1.2)^2 = 1.518 \text{ mA}
 \end{aligned}$$

(b) Find the small-signal voltage gain for $R_L = \infty$:

$$\begin{aligned}
 g_m &= 2\sqrt{K_p I_{DQ}} = 2\sqrt{(2)(1.518)} = 3.485 \text{ mA/V} \\
 r_o &= \frac{1}{\lambda I_{DQ}} = \frac{1}{(0.03)(1.518)} = 22 \text{ k}\Omega \\
 A_v &= -g_m(R_D \parallel r_o) = -(3.485)(4 \parallel 22) = -11.8
 \end{aligned}$$

(c) Repeat part (b) for $R_L = 8$ k Ω :

$$A_v = -g_m(R_D \parallel r_o \parallel R_L) = -(3.485)(4 \parallel 22 \parallel 8) = -8.29$$

Problem 4.36

The parameters of the circuit in Figure P4.36 are $R_S = 4 \text{ k}\Omega$, $R_1 = 850 \text{ k}\Omega$, $R_2 = 350 \text{ k}\Omega$, and $R_L = 4 \text{ k}\Omega$. The transistor parameters are $V_{TP} = -1.2 \text{ V}$, $k'_p = 40 \text{ }\mu\text{A/V}^2$, $W/L = 80$, and $\lambda = 0.05 \text{ V}^{-1}$. (a) Determine I_{DQ} and V_{SDQ} . (b) Find the small-signal voltage gain $A_v = v_o/v_i$. (c) Determine the small-signal circuit transconductance gain $A_g = i_o/v_i$. (d) Find the small-signal output resistance R_o .

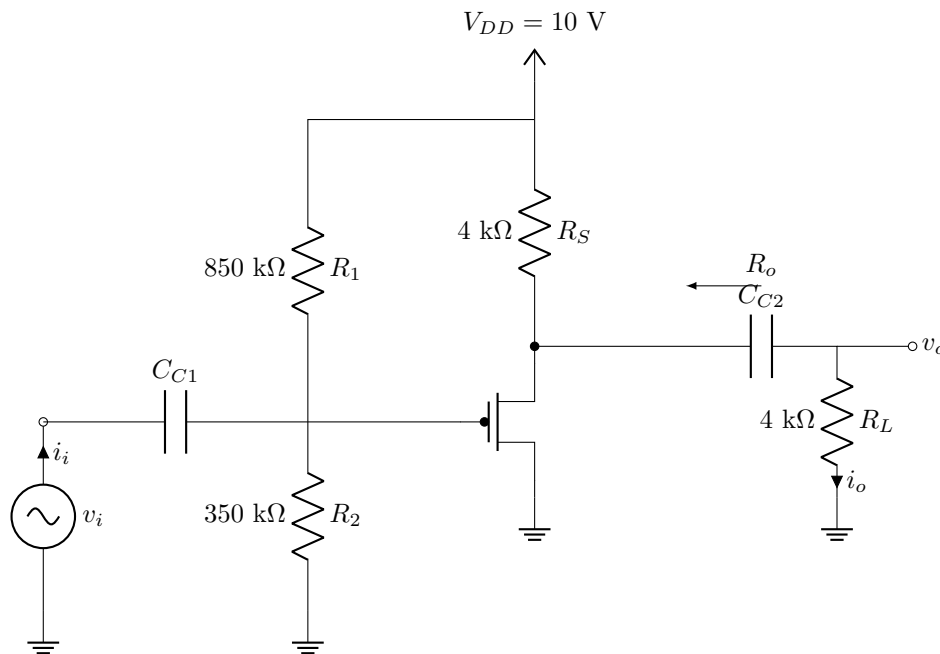


Figure 4: Figure P4.36

Solution

(a) Determine I_{DQ} and V_{SDQ} :

$$\begin{aligned}
 V_G &= \left(\frac{R_2}{R_1 + R_2} \right) V_{DD} = \left(\frac{350}{350 + 850} \right) (10) = 2.917 \text{ V} \\
 V_{DD} &= I_{DQ} R_S + V_{SGQ} + V_G \Rightarrow 10 = I_{DQ}(4) + V_{SGQ} + 2.917 \\
 K_p &= \frac{1}{2} k'_p \frac{W}{L} = \frac{1}{2} (0.04 \text{ mA/V}^2) (80) = 1.6 \text{ mA/V}^2 \\
 10 &= (1.6)(V_{SGQ} - 1.2)^2 (4) + V_{SGQ} + 2.917 \\
 10 &= 6.4(V_{SGQ}^2 - 2.4V_{SGQ} + 1.44) + V_{SGQ} + 2.917 \\
 0 &= 6.4V_{SGQ}^2 - 14.36V_{SGQ} + 2.133 \Rightarrow V_{SGQ} = 2.084 \text{ V} \\
 I_{DQ} &= 1.6(2.084 - 1.2)^2 = 1.25 \text{ mA} \\
 V_{SDQ} &= V_S - V_D = (10 - I_{DQ} R_S) - 0 = 10 - (1.25)(4) = 5 \text{ V}
 \end{aligned}$$

(b) Find the small-signal voltage gain $A_v = v_o/v_i$:

$$\begin{aligned}
 g_m &= 2\sqrt{K_p I_{DQ}} = 2\sqrt{(1.6)(1.25)} = 2.828 \text{ mA/V} \\
 r_o &= \frac{1}{\lambda I_{DQ}} = \frac{1}{(0.05)(1.25)} = 16 \text{ k}\Omega \\
 A_v &= \frac{g_m(r_o \parallel R_S \parallel R_L)}{1 + g_m(r_o \parallel R_S \parallel R_L)} \\
 r_o \parallel R_S \parallel R_L &= 16 \parallel 4 \parallel 4 = 1.778 \text{ k}\Omega \\
 A_v &= \frac{(2.828)(1.778)}{1 + (2.828)(1.778)} = 0.834
 \end{aligned}$$

(c) Determine the small-signal circuit transconductance gain $A_g = i_o/v_i$:

$$A_g = \frac{i_o}{v_i} = \frac{i_o}{v_o} \frac{v_o}{v_i} = \frac{1}{R_L} A_v = \frac{1}{4} (0.834) = 0.2085 \text{ mA/V}$$

(d) Find the small-signal output resistance R_o :

$$\begin{aligned}
 R_o &= \frac{1}{g_m} \parallel R_S \parallel r_o \\
 \frac{1}{R_o} &= g_m + \frac{1}{R_S} + \frac{1}{r_o} = 2.828 + \frac{1}{4} + \frac{1}{16} = 3.1405 \text{ mS} \\
 R_o &= \frac{1}{3.1405} = 0.3184 \text{ k}\Omega = 318.4 \text{ }\Omega
 \end{aligned}$$